

Arnav Shukla

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Summary

Full-stack and applied-AI engineer building production-grade systems, from row-level-secured healthcare platforms to autonomous, fault-tolerant AI agents and resource-constrained ML pipelines. End-to-end system development: database design, backend APIs, and deployment. Benchmarks performance, latency, memory, and security.

Technical Skills

Languages	Python, Java, C/C++, JavaScript, TypeScript, SQL
Frontend	React, Next.js, Vue.js, HTML5, CSS3, Tailwind CSS, Zod
Backend	FastAPI, Flask, Django, Node.js, REST APIs, WebSockets
Databases	PostgreSQL, MySQL, SQLite, Redis, Supabase, PostGIS, SQLAlchemy, Row-Level Security (RLS)
Cloud / DevOps	AWS (Cloud Architecting), Docker, Docker Compose, Vercel, Render
AI / ML	Gemini, Groq, Ollama, Hugging Face, scikit-learn, Whisper, spaCy, Prompt Engineering, RAG, MCP
Developer Tools	Git, GitHub, Postman, Playwright, VS Code, Figma

Projects

- Rural Healthcare Platform** | *Next.js, TypeScript, Supabase, PostgreSQL, PostGIS, Gemini* | Github · Live
- Built a full-stack bilingual (English/Hindi) telehealth platform with Row-Level Security across **11 PostgreSQL tables**; verified SQL-injection and RLS-bypass attempts return **0 rows** in production
 - Shipped a Gemini-powered symptom checker and PostGIS `ST_DWithin` geo-search over **6,616 indexed facilities**, averaging **274ms** query latency (P95 558ms) via a GIST spatial index
 - Designed 12 rate-limited REST routes with Zod validation and Jitsi teleconsultation; achieved **100% pass rate** on a 10-point automated security audit (HSTS, CSP, SAMEORIGIN) and **197ms average TTFB** on Vercel Edge (Mumbai)
- Eco-Loop: Autonomous Building AI** | *FastAPI, React, Ollama/Qwen2.5, FastMCP* | Github · Honeywell Hackathon
- Developed an autonomous HVAC control agent using a locally-hosted Qwen2.5 (1.5B) LLM via Ollama, cutting simulated energy consumption by **50%** (0.86 vs. 1.71 kWh) against a rule-based baseline while maintaining ASHRAE 55 comfort (avg PMV 0.00) across a 5-zone office
 - Exposed a **9-tool** Model Context Protocol (MCP) server (avg **0.006ms** execution, 100% success rate) so any MCP-compliant AI client can control the simulated building
 - Built a fault-tolerant control loop with automatic rule-based fallback, sustaining **100% uptime with zero crashes** across 30 injected disconnect, timeout, and malformed-response failures, streamed over WebSockets to React dashboard
- TalentMatch AI v2** | *Python, FastAPI, C++, sentence-transformers, Docker* | Github · Live
- Architected a resume-scoring pipeline combining a custom-compiled C++ scoring engine with sentence-transformer embeddings (`all-MiniLM-L6-v2`), averaging 4.7s end-to-end inference (PDF parse + embedding + scoring) under load
 - Engineered a multi-stage Docker pipeline that compiles the native C++ engine and pre-bakes the ML model at build time, eliminating cold-start model downloads; hardened with fully environment-isolated secrets and graceful error handling on malformed uploads (clean HTTP 400 instead of a crash)
 - Shipped a 21.8KB vanilla-JS Chrome extension (zero framework overhead) that extracts job descriptions from any page and feeds them directly into the scoring pipeline
- CampusCart: Multi-Vendor Campus Marketplace** | *Python, Flask, SQLAlchemy, SQLite* | Github · Live
- Engineered a multi-tenant marketplace across **3 roles** (Student, Vendor, Admin) using Flask Blueprints, with catalog queries executing in **under 1ms** and **0% error rate** across benchmark testing
 - Designed a relational schema with strict foreign-key constraints, Bcrypt-hashed authentication, role-based access decorators, and a simulated campus wallet, running on **under 85MB RAM**

Education

Vellore Institute of Technology - Bhopal <i>Bachelor of Technology, Computer Science and Engineering; CGPA: 8.6</i> Relevant Coursework: DSA, OOP, OS, DBMS, Computer Networks	May 2027 Bhopal, MP, India
Higher Secondary (M.P. Board) – 83.4%	May 2023
Secondary (CBSE) – 94.33%	May 2021

Certifications

ServiceNow Virtual Internship Program – ServiceNow University, AICTE, SmartBridge	Jul 2026
AWS Academy Graduate – Cloud Architecting (60 hrs)	Jun 2026
Machine Learning – NPTEL/IIT Madras Marketing Analytics – NPTEL/IIT Kharagpur	2025–26
AI Knowledge Challenge (Winner) – IBM Call for Code Computer Networking – Coursera/Google	2024

Extracurricular Activities

- Contested for Honeywell Hackathon (Eco-Loop) round 1; pitched HealthSeva AI at Smart India Hackathon 2025 (college round); active participant in VIT Bhopal tech fest